

Problem 1: Two players alternate taking 1, 3, or 4 counters. Taking 2 is not allowed. Never take more than remain. Taking the last counter wins. Play from 5, 6, and 7. Swap who starts and replay each pile. Record the moves in order; a partner may write while you play.

Starting pile	First move	Next moves, in order	Winner: first or second
5			
5			
6			
6			
7			
7			

Problem 2: Build a pile of 2, then a pile of 4. From each pile, try every allowed first move. Reset the pile before the next attempt. Record how many counters each move leaves.

Pile	Allowed move / counters left
2	
4	

Problem 3: Use moves 1, 3, and 4. A position is W if the player about to move has a winning strategy; it is L if every move lets the other player force a win. Put L at 0 because there is no move. Work from small piles upward. List all legal next piles. Mark W if at least one is L; mark L if all are W.

Pile	All legal next piles	W or L	One move to L, if any
0	none	L	none
1			
2			
3			
4			
5			
6			
7			

Problem 4: Build a nonempty pile you marked L, such as 2 or 7. Let your partner choose a legal move, then use your chart to reply. Reset and try every different first move. Record your replies.

Starting pile, each first move, and my replies:

Problem 5: Use moves 1, 3, and 4. From 8 counters, try taking 1, 3, and 4. Reset each time. Use your chart to record what each move leaves and whether that next pile is W or L. Circle every winning first move.

Take	Pile left	Its W/L label
1		
3		
4		

Problem 6: Do the same from 10. First decide the missing labels for 8 and 9, using all their legal moves. Record all winning moves from 10, not just the first one you find.

Take	Pile left	Its W/L label
1		
3		
4		

Problem 7: Choose a starting pile from 8 through 14. Make a strategy card: record each possible first move and the pile it leaves. Circle moves that hand your opponent an L. Test the card in a game.

My pile and strategy card: